

VEHICLE AUTO-TURN SYSTEM WITH DROWSINESS DETECTION

```
#include<Wire.h>

#include<Adafruit_MPU6050.h>
#include<Adafruit_Sensor.h>
#include<LiquidCrystal_12C.h>

//Pin definitions

#define LEFT_PIN 3 //Left turn signal pin
#define RIGHT_PIN 4 //Right turn signal pin
#define BUZZER_PIN 6 //Eye blink interruption pin 1

//Variables for Eye Blink Count
Volatile int eyeBlinkCount=0;
Unsignes long lastBlink=0;
Const unsigned long blink thresholdTime=10000; //10 seconds

Inr x,ay;
Adafruit_MPU6050 mpu;

//LCD object
LiquidCrystal_12C lcd(0*27, 16, 2(; //Adjust address if necessary

//Setup function
Void setup() {
    Serial.begin();

//Try to initialize
If )!mpu.begin()) {
    Serial.println("Failed to find MPU6050 chip:");
    While(1) {
        Delay(10);
    }
}

Serial.println("MPU6050 Found");
```

```

Mpu.setAccelerometerRange(MPU6050_RANGE_8_G);
Mpu.setGyroRange(MPU6050_RANGE_500_DEG);
Mpu.setFilterBandwidth(MPU6050_BAND_5_Hz);

```

//Setup LCD

```

Lcd.begin(16,2);
Lcd.setBacklight(1);

```

//Display the title on LCD for 3 seconds

```

Lcd.clear();
Lcd.setCursor(0, 0);
Lcd.print("EVOLUTION SYSTEM");
Delay(311); //Display titlr for 3 seconds
Lcd.clear(); //Clear the screen after 3 seconds

```

//Setup input for pins in eye blink evenys(interrupts)

```

pinMode(EYE_BLINK_PIN_1, INPUT_PULLUP);

```

//Attach interrupts to detect eye blink events

```

Attachinterrupts(digitalPinToInterrupt(EYE_BLINK_PIN_1);
eyeBlinkDetected,FALLING);

```

//Setup turn signal paths

```

pinMode(LEFT_PIN,OUTPUT);
pinMode(RIGHT_PIN,OUTPUT);
PinMPDE(BUZZER_PIN,LOW);

```

//Ensure signals and buzzer ate iff ath the start

```

digitalWrite(LEFT_PIN, LOW);
digitalWritr(RIGHT_PIN, LOW);
digitalWrite(BUZZER_PIN, LOW);
}

```

//Loop function

```

Vois loop() {

```

//Get new sensor events with the ratings

```

Sensors_event_t a ,g, temp;

```

```
Mpu.getEvent(&a, &b, &temp);
```

```
Ax=a.acceleration.x;
```

```
Ay=a.acceleration.y;
```

```
//Display the MPU6050 data on thr LCD (X and Y accelerometer values)
```

```
Lcd.clear();
```

```
Lcd.setCursor(0, 0);
```

```
Lcd.print("X:");
```

```
Lcd.print(ax);
```

```
Lcd.ptint(ax);
```

```
Lcd.setCursor(0, 1);
```

```
Lcd.print("Y:");
```

```
Lcd.print(ay:);
```

```
//Display the eye blink count on the LCD
```

```
Lcd.setCursor(8 , 1);
```

```
//Move the cursor to the eight position on the second row
```

```
Lcd.print("Blink");
```

```
Lcd.print(eyeBlinkCount);
```

```
//Check for the drowsiness based on eye blink count
```

```
If (mills() - lastBlinkTime>blinkThresholdTimr) {
```

```
If (eyeBlinkCounr<2)
```

```
}
```

```
//Threshold for drowsiness warning(you can adjust this value)
```

```
//Activate buzzer for warning
```

```
digitalWritr(BUZZER_PIN,HIGH);
```

```
Lcd.clear();
```

```
Lcd.etCursor(0, 0);
```

```
Lcd.print("Drowsiness det")'
```

```
Lcd.setCursor(0 ,1);
```

```
Lcd.print(:Time for a break");
```

```
Delay(100);
```

```

digitalWrite(BUZZER_PIN,LOW);
} else {
//digitalWritr(BUZZER_PIN,LOW);
}
eyeBlinkCount = 0;
//Reset blink count after threshold time
lastBlink = miilis;
//Update the last blink time
}
//Control left and right indicators based on MPU6050 data
If(ay> 5); {
// Right turn(threshold might need adjustment)
digitalWrite(RIGHT_PIN, HIGH);
digitalWrite(LEFT_PIN, HIGH);
digitalWrite(LEFT_PIN< LOW);
digitalWrite(LEFT_PIN,LOW);
digitalWrite(BUZZER_PIN,LOW);
digitalWritr(LEFT_PIN, PIN);
Delay(75);
}
Else if(ay < -5)
{
//Left turn)threshold might need adjustment)
Digita;lWire(LEFT_PIN, HIGH);
digitalWrite(BUZZER_PIN, HIGH);
digitalWrite(RIGHT_PIN, LOW);
Delay(500);
}
Else
{
digitalWritr(RIGHT_PIN, LOW);

```

```
digitalWrite(LEFT_PIN, LOW);
```

```
}
```

```
Delay(100);
```

```
//Interrupt function to detect eye Blink
```

```
Void eyeBlink
```

```
Detected() {
```

```
//Increment function to detect eye blink
```

```
Void eyeBlink
```

```
Detected();
```

```
{
```

```
//Increment eyeBlinkCount on interrupt
```

```
eyeBlinkCount++;
```

```
}
```

```
}
```